

8 (short)

(a) Describe with sketches how substitutional solid solutions provide pinning points to dislocations, increasing the yield stress. [4]

(b) The Cu-Ni system forms solid solutions for all compositions between pure Cu and pure Ni. Figure 8 shows the variation of the yield stress with composition of Ni in weight %. For low concentrations (< 20%), the yield stress σ_y is expected to vary with composition C as

$$\sigma_y = \sigma_0 + \alpha C^n \text{ MPa}$$

where α and n are constants.

(i) Define the physical meaning of σ_0 . [2]

(ii) Use the graph to find the constants α and n for dilute solutions of Ni in Cu, with stress σ in MPa and composition C in wt%. [2]

(iii) Assuming the value of n is the same for dilute solutions of Cu in Ni, state by inspection whether the constant α is higher or lower in this case. [2]

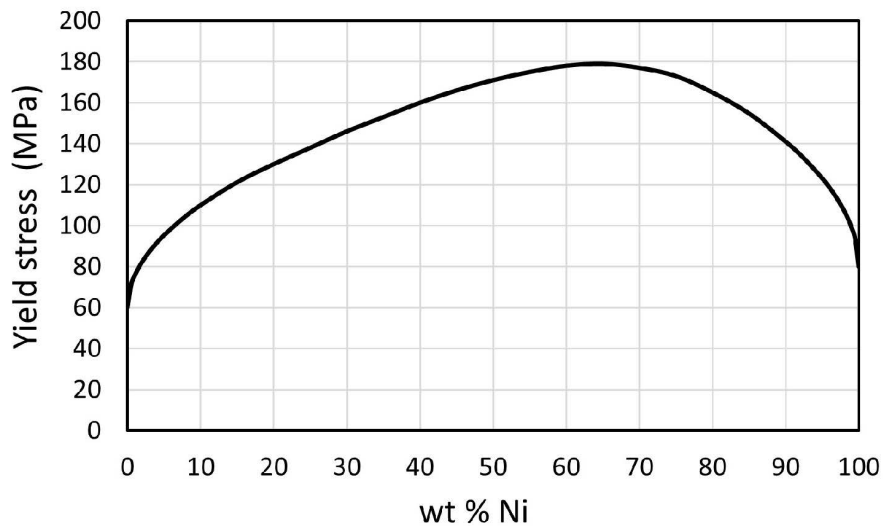


Fig. 8